



Microsoft Azure Fundamentals

Regional School of DevOps

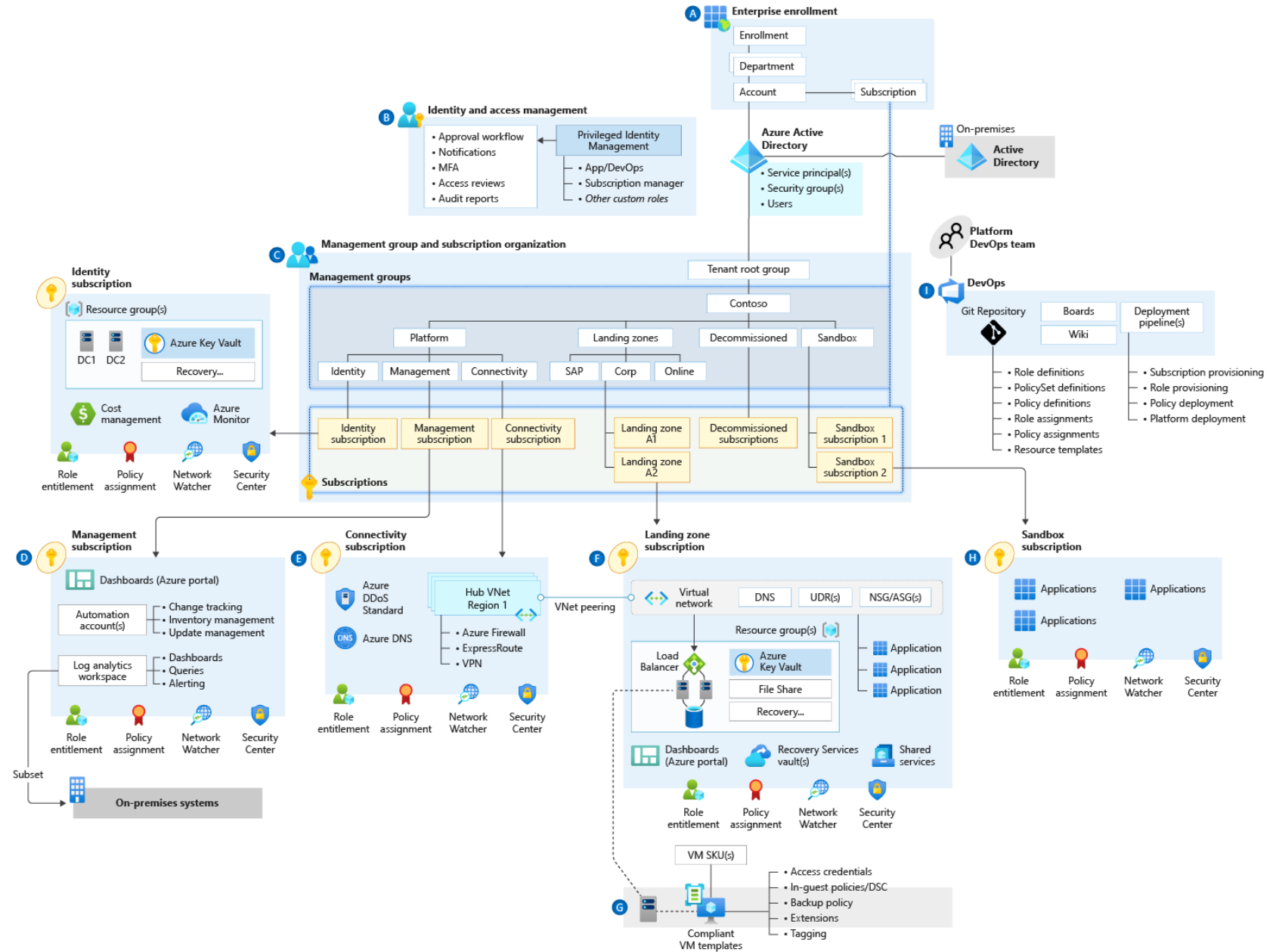


Agenda

1. Core Azure architectural components
2. Compute services
3. Network services
4. Storage and Database services
5. Monitoring
6. Security
7. Azure Resource Manager



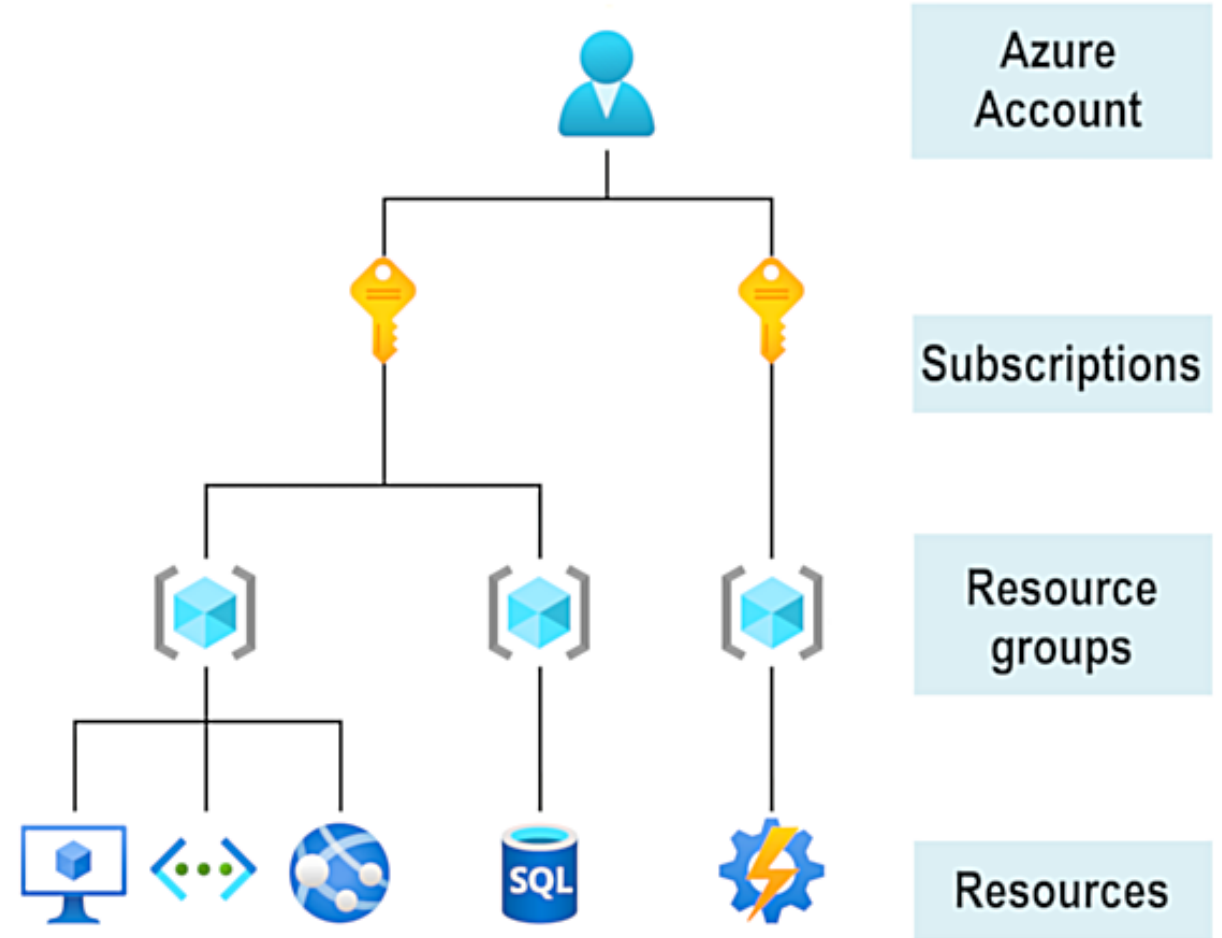
Azure Tenants



Azure subscription

- To create Azure services, a subscription is needed.
- An account can be assigned to multiple subscriptions:
 - PROD
 - NON-PROD
- Resource costs differ from resource to resource.
- Pay-as-you-go.

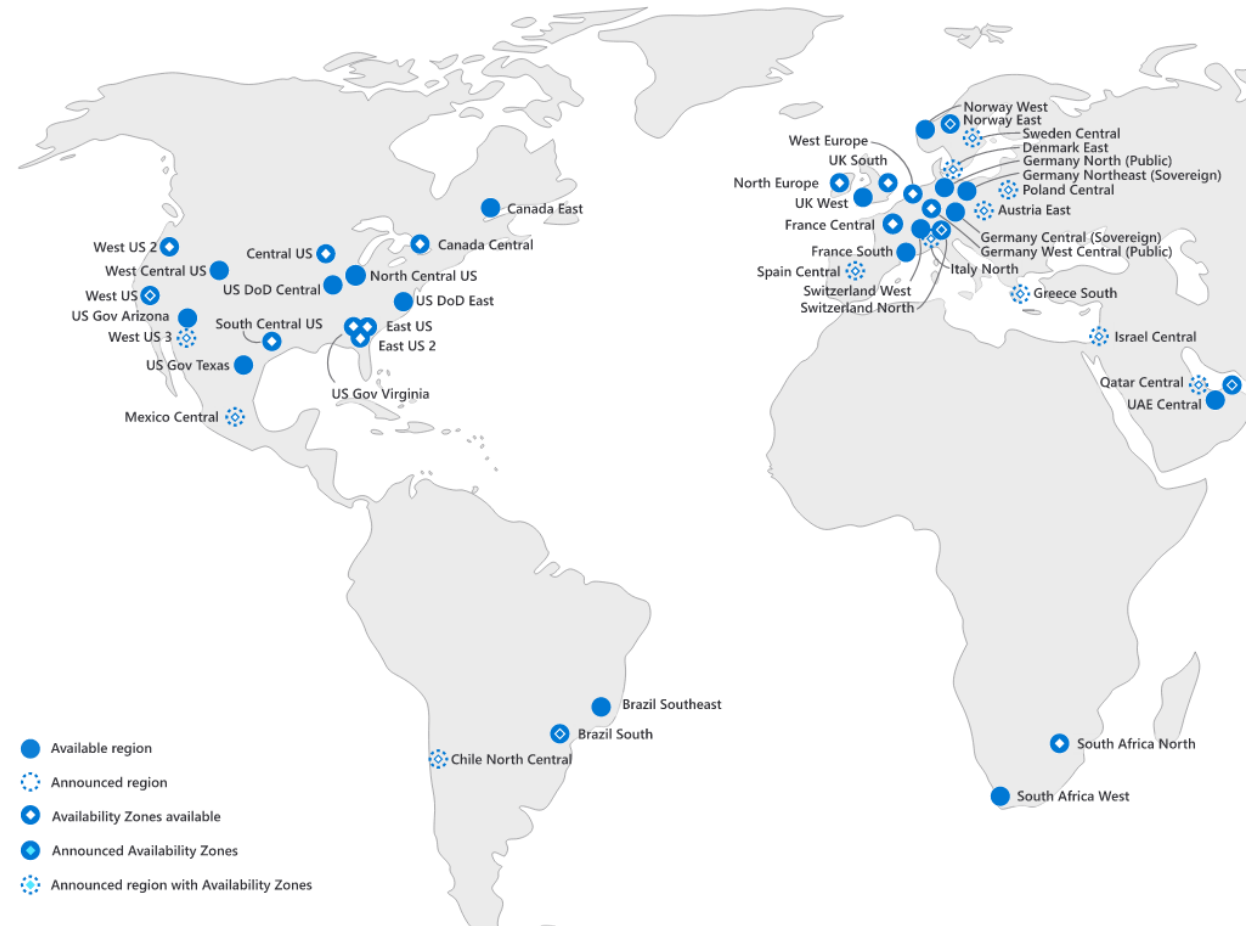
[Azure Pricing Calculator](#)

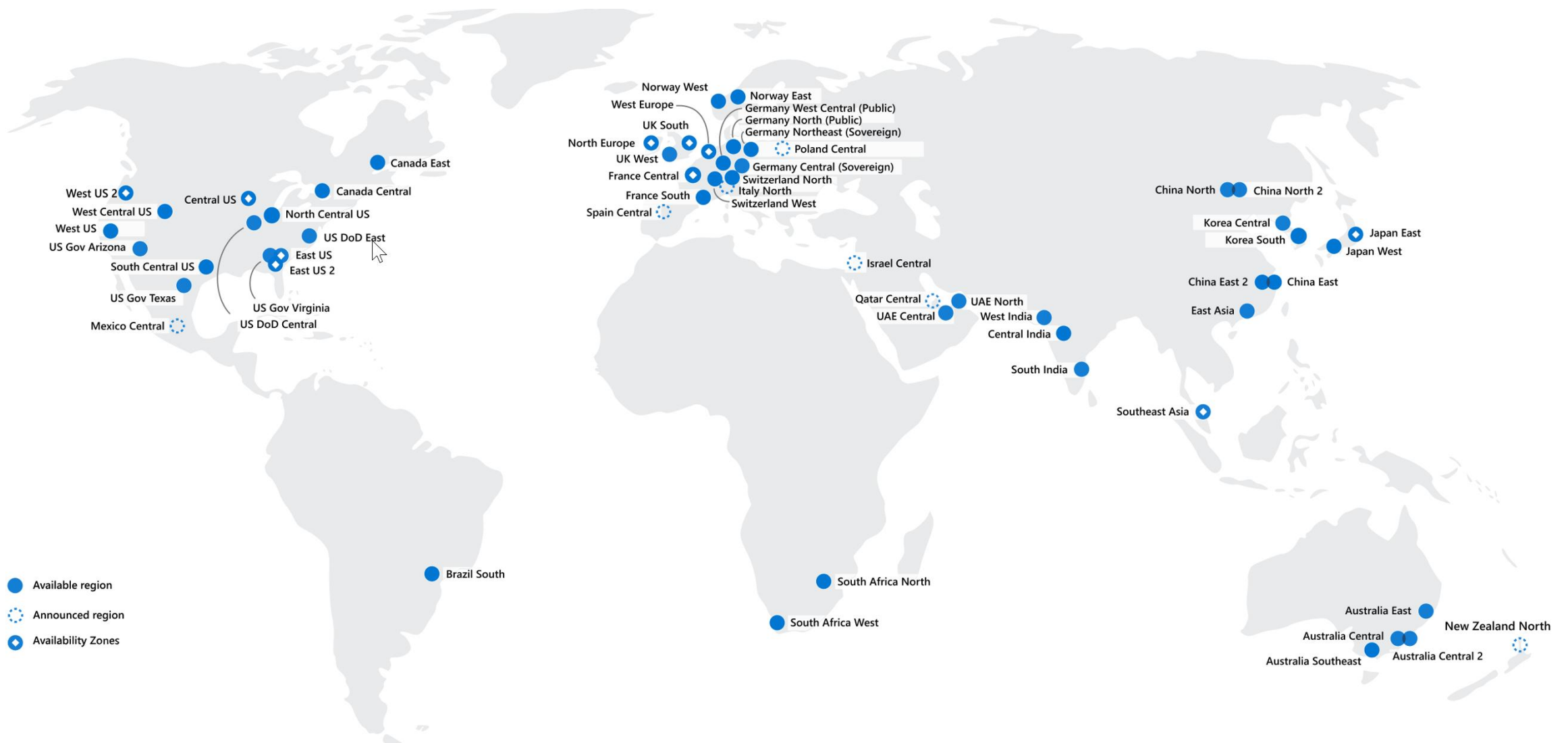


Regions

- A region represents a collection of datacenters.
- Preserves data residency.
- Select regions close to the users.
- There are global services that are region dependent
- Special Azure regions:
 - US Gov and US DoD
 - China East and North

[Azure Speed Test](#)





Azure regions as of June 2020

Geographies and region pairs

- Geographies are discrete markets that preserve data residency and compliance boundaries.
- Typically contain two or more regions.
- Americas, Europe, Asia Pacific, Middle East and Africa
- Each Azure region is paired within another region within the same geography.

Region		Region
North Central US		South Central US
East US		West US
West US 2		West Central US
US East 2		Central US
Canada Central		Canada East
North Europe		West Europe
UK West		UK South
Germany Central		Germany Northeast
South East Asia		East Asia
East China		North China
Japan East		Japan West
Australia Southeast		Australia East
India South		India Central
Brazil South (Primary)		South Central US

Security & Management

-  Security Center
-  Azure portal
-  Azure Active Directory
-  Azure AD B2C
-  Multi-Factor Authentication
-  Automation
-  Key Vault
-  Azure Marketplace
-  VM Image Gallery
-  REST API and CLI

Platform Services

Media & CDN

-  Media Services
-  Media Analytics
-  Content Delivery Network

Integration

-  API Management
-  Service Bus
-  Azure Logic Apps

Compute Services

-  Container Service
-  VM Scale Sets
-  Azure Batch
-  Dev/Test Lab

Application Platform

-  Web Apps
-  Mobile Apps
-  API Apps
-  Cloud Services
-  Service Fabric
-  Notification Hubs
-  Functions

Developer Services

-  Visual Studio
-  Mobile Engagement
-  Azure DevOps
-  Xamarin
-  Application Insights
-  Visual Studio App Center

Data

-  SQL Database
-  Azure Synapse Analytics
-  Cosmos DB
-  SQL Server Stretch Database
-  Azure Cache for Redis
-  Table Storage
-  Azure Search

Intelligence

-  Cognitive Services
-  Bot Services
-  Azure ML Studio

Analytics & IoT

-  HDInsight
-  Machine Learning
-  Stream Analytics
-  Data Catalog
-  Data Lake Analytics Service
-  Data Lake Storage
-  IoT Hub
-  Event Hubs
-  Data Factory
-  Power BI Embedded

Hybrid Cloud

-  Azure AD Connect Health
-  AD Privileged Identity Management
-  Domain Services
-  Backup
-  Azure Monitor
-  Import/Export
-  Azure Site Recovery
-  StorSimple

Infrastructure Services

Compute

-  Virtual Machines
-  Containers and Azure Kubernetes

Storage

-  Blob
-  Queues
-  Files
-  Disks

Networking

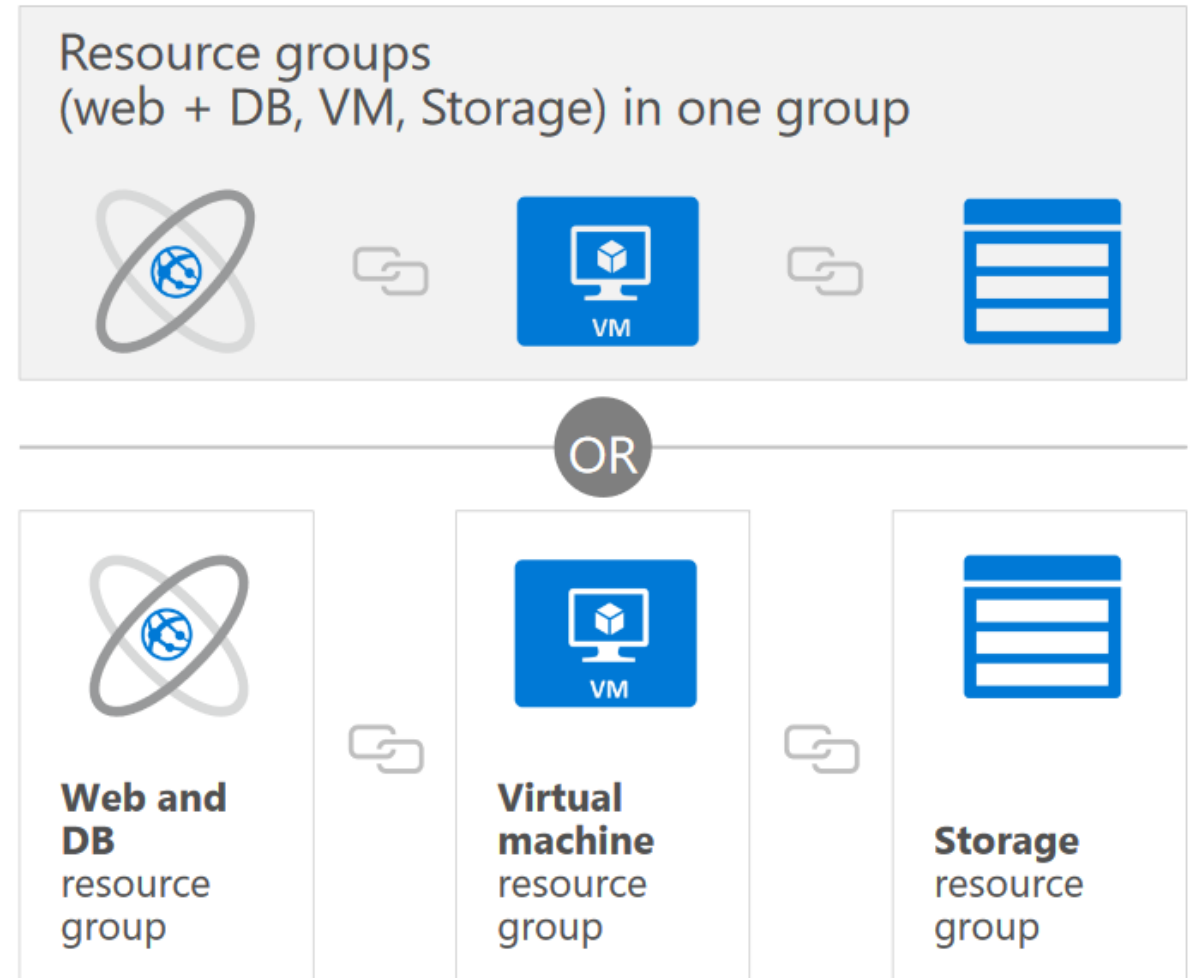
-  Virtual Network
-  Load Balancer
-  DNS
-  Express Route
-  Traffic Manager
-  VPN Gateway
-  App Gateway

Datacenter Infrastructure



Resource groups

- Logical containers used for organizing Azure resources.
- Every Azure resource must exist in only one resource group.
- Highly recommended to create the resource group in the same region as the resources.
- Secure at the resource (group) level using RBAC (Role-Based Access Control).



Agenda

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2. Compute services
3. Network services
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6. Security
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Azure compute services



Azure VMs



Containers

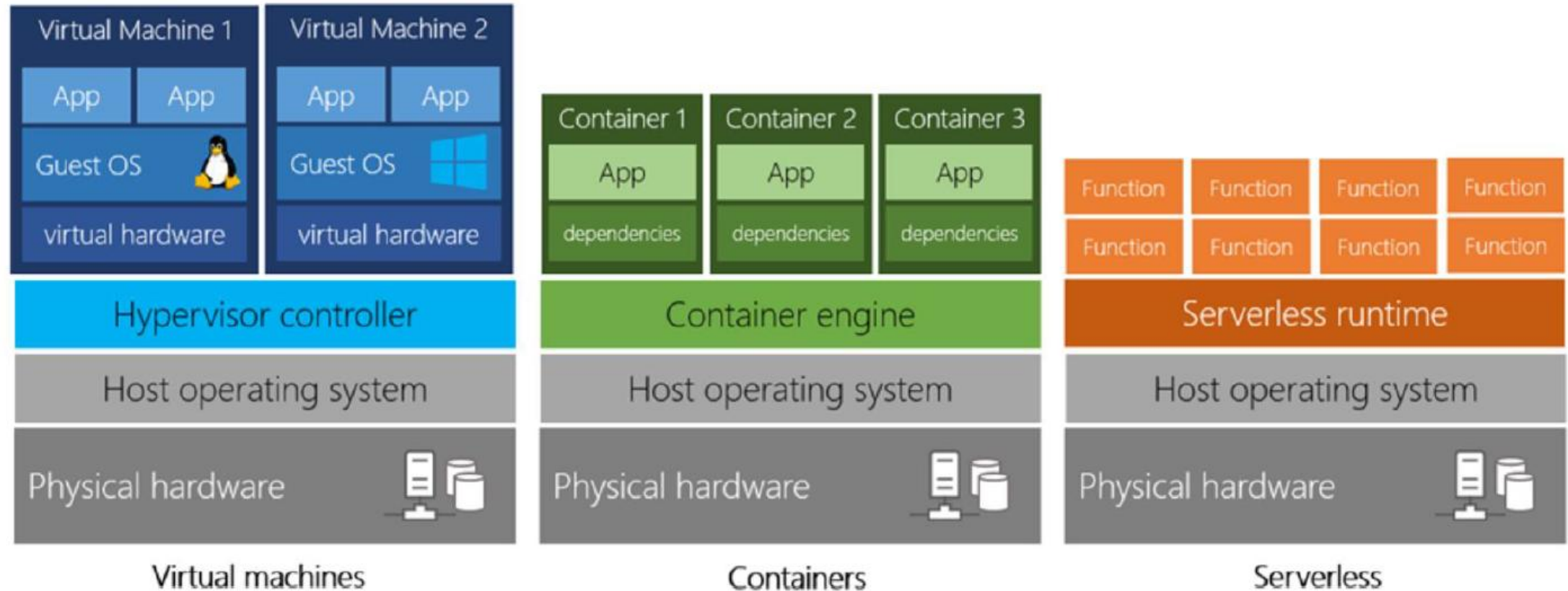


App services



Serverless

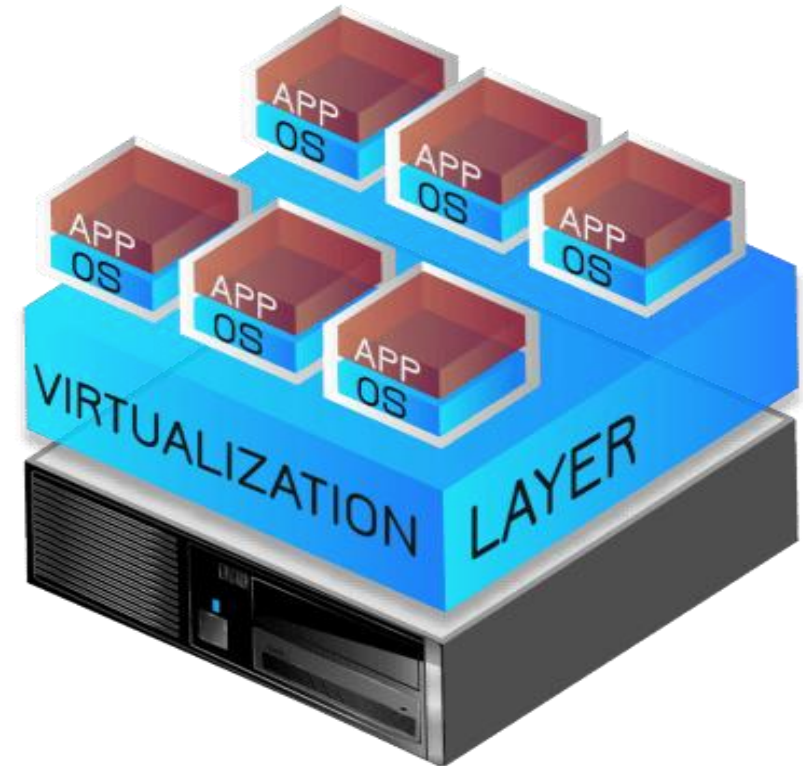
Comparing compute options



Virtual Machines



Traditional Architecture



Virtual Architecture

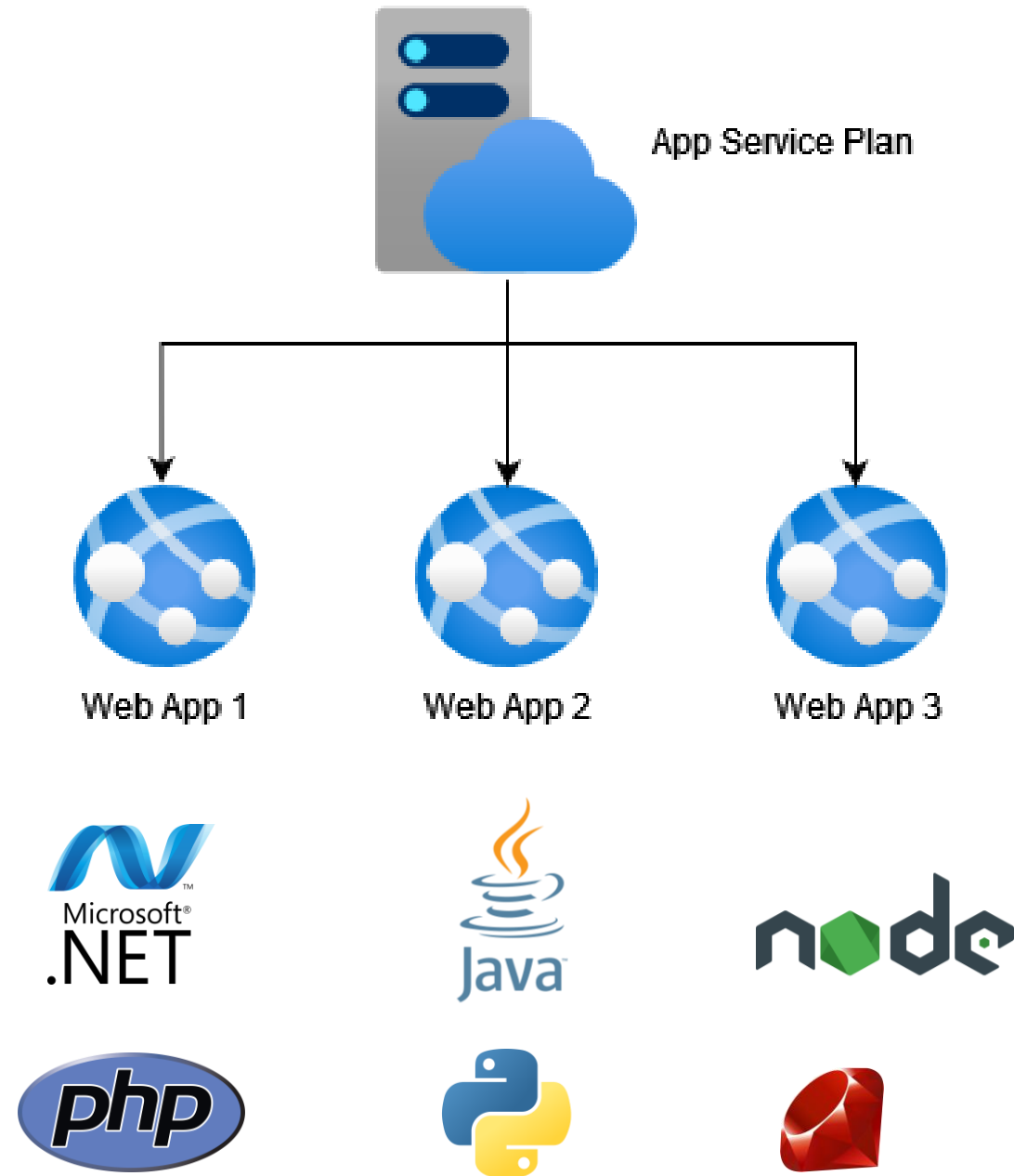
Azure VM types



	General Purpose	Compute Optimized	Memory Optimized	Storage Optimized	GPU	High Performance
Type	A, B, Dsv3, Dv3, DaSv4, Dav4, DSv2, Dv2, Av2, DC, DCv2	Fsv2, Fs, F	Esv3, Ev3, Easv4, Eav4, Mv2, M, DSv2, Dv2	Lsv2	NC, NCv2, NCv3, ND, NDv2 (Preview), NV, NVv3, NVv4	HB, HBv2, HC, H
Description	Balanced CPU-to-memory ratio.	High CPU-to-memory ratio	High memory-to-CPU ratio	High disk throughput and IO	GPU based VMs with single and multiple GPUs	Most powerful CPU virtual machines
Usage	Ideal for testing and development. Small to medium databases. Low to medium traffic web servers.	Medium traffic web servers. Network appliances. Batch processes, and application servers	Relational database servers. Medium to large caches. In-memory analytics.	Ideal for Big Data, SQL NoSQL databases. Data warehousing. Large transactional databases	Heavy graphics rendering and video editing. Model training and inferencing (ND) with deep learning.	Optimized for applications driven by memory bandwidth, such as fluid dynamics, explicit finite element analysis, and weather modelling.

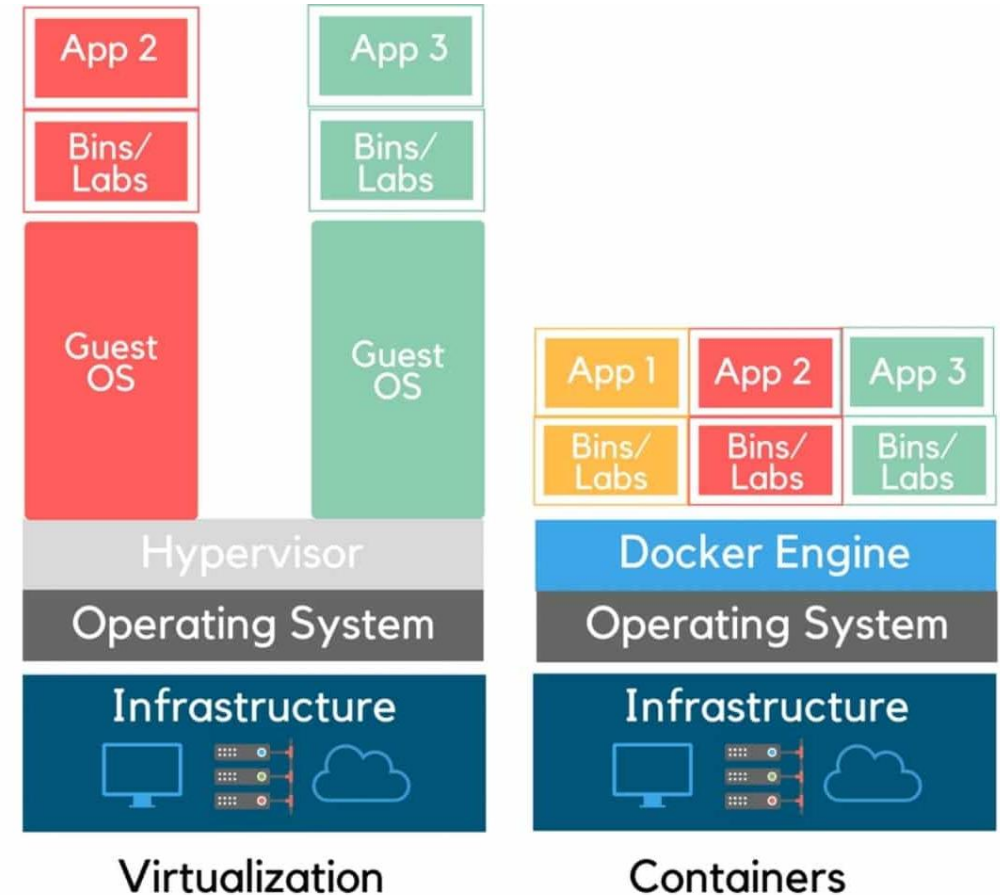
Azure App Services

- PaaS offer of compute services.
- Supports both Linux and Windows.
- Apps always runs in an App Service Plan.
- App Service Plan defines the set of resources used.
- Has built-in monitoring.



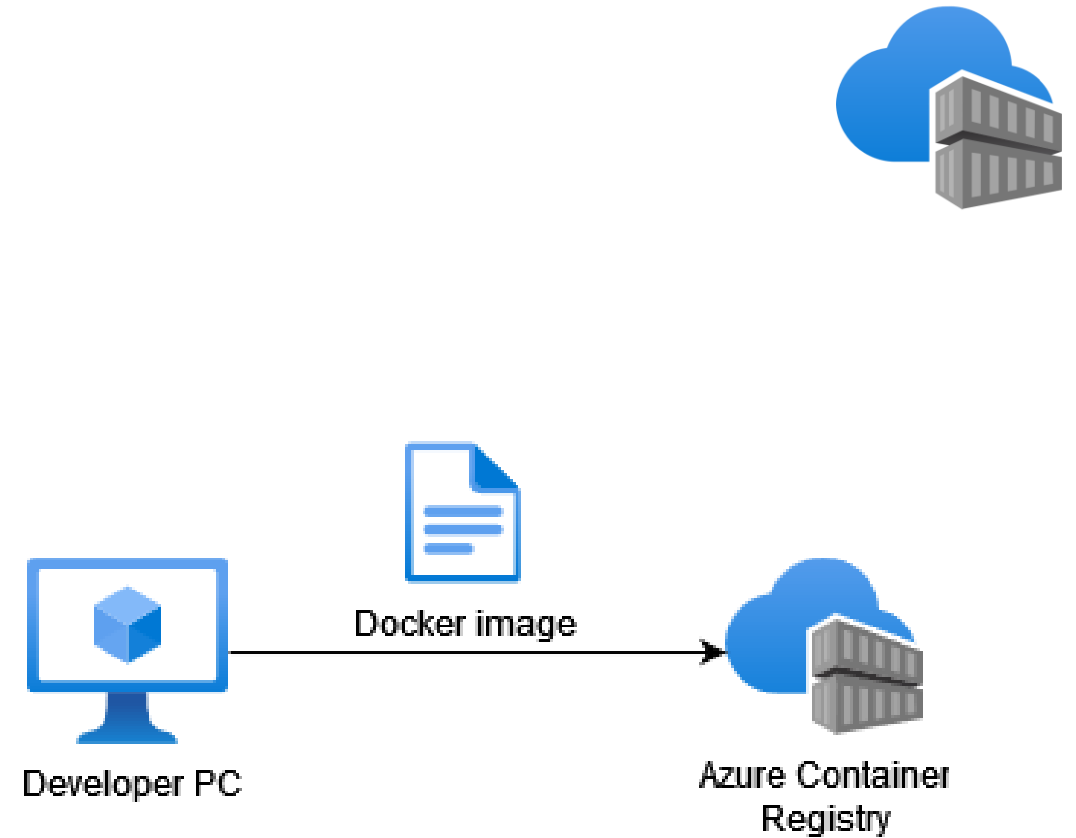
Container services

- Containers are virtualization environments.
- Unlike VMs, you don't manage an OS.
- Are meant to be lightweight, and to be created, scaled and stopped dynamically.



Azure Container Registry

- Store and manage your private Docker container images and related artifacts.
- Integration with Azure DevOps and Jenkins.
- Highly secured:
 - image transfer over HTTPS
 - client connection via TLS



Azure Container Instance

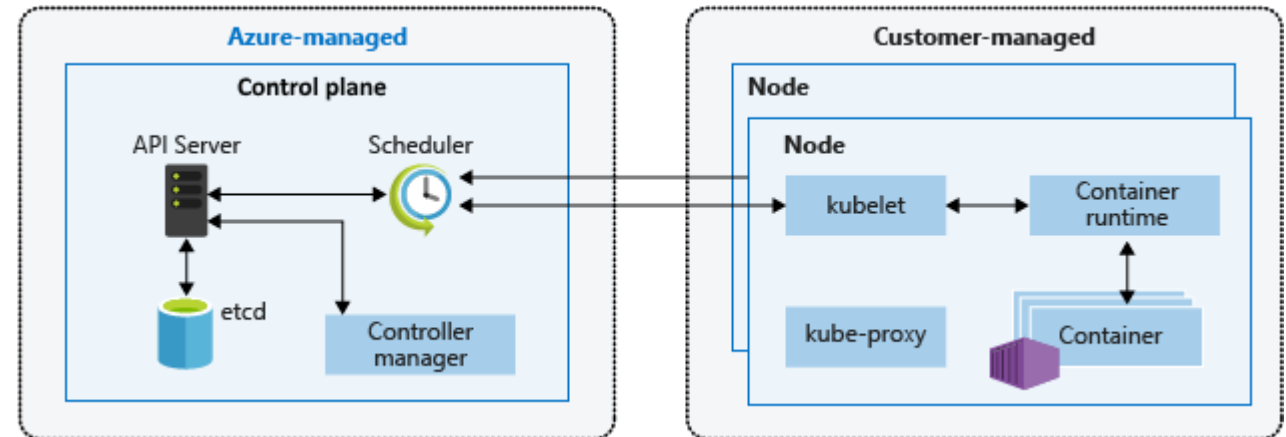
- Hosts and runs your containers.
- Supports Linux and Windows containers.
- Integration with Azure Container Registry and Docker Hub.
- Can be directly exposed to the internet via a public IP.
- Supports custom DNS name:
customlabel.azureregion.azurecontainer.io



Azure Kubernetes Service



- A container orchestrator service for managing large numbers of containers.
- Pay for the agent nodes within the cluster.
- Two main components:
 - Control plane (Azure managed).
 - Nodes (User managed).



Serverless computing

- There are still servers!
- Fully-managed services.
- Only pay for what you use.
- Flexibility to scale as needed.
- Serverless technologies work between them.



Serverless computing services



- **Azure Functions:** serverless compute resource that runs your code. Build once, deploy anywhere.

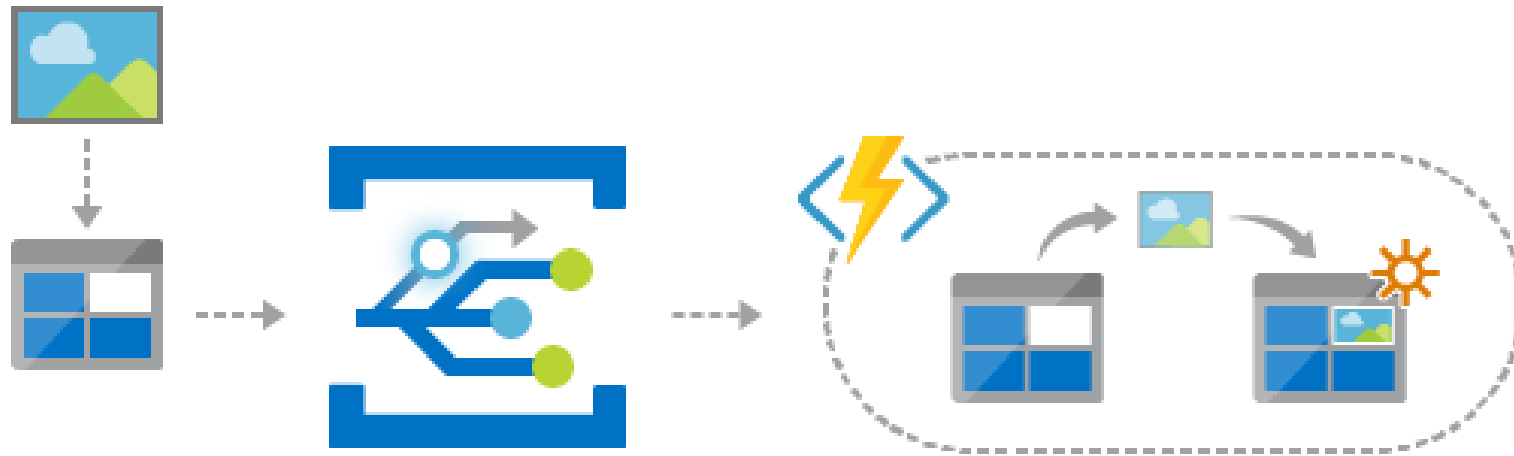


- **Azure Logic Apps:** schedule, automate, and orchestrate tasks, business processes, and workflows when you need to integrate apps, data, systems, and services.



- **Event Grid:** build applications with event-based architectures.

Serverless app architecture



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Azure network services



Virtual Networks



VPN Gateway



**Content Delivery
Network**



Azure DNS



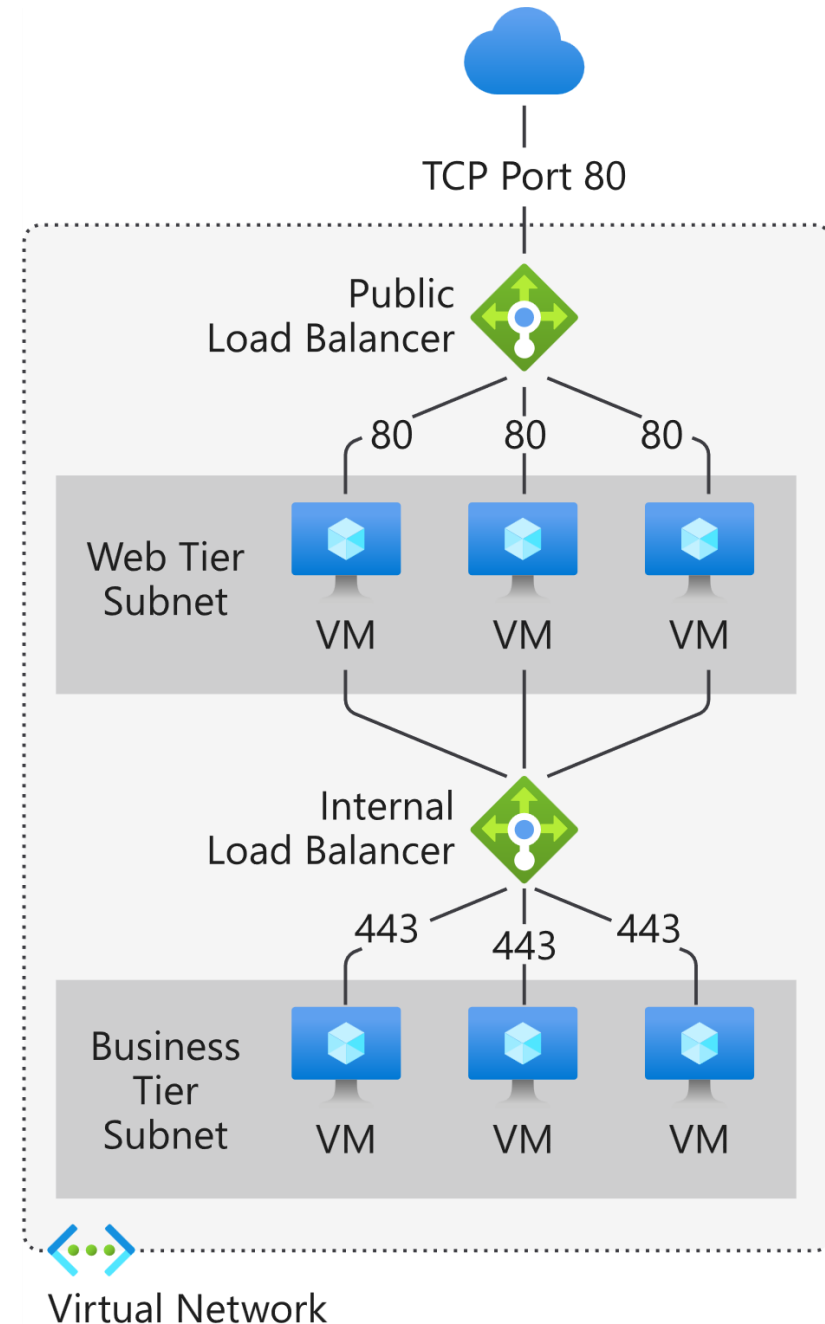
Private Endpoint



**Network Interface
Card**

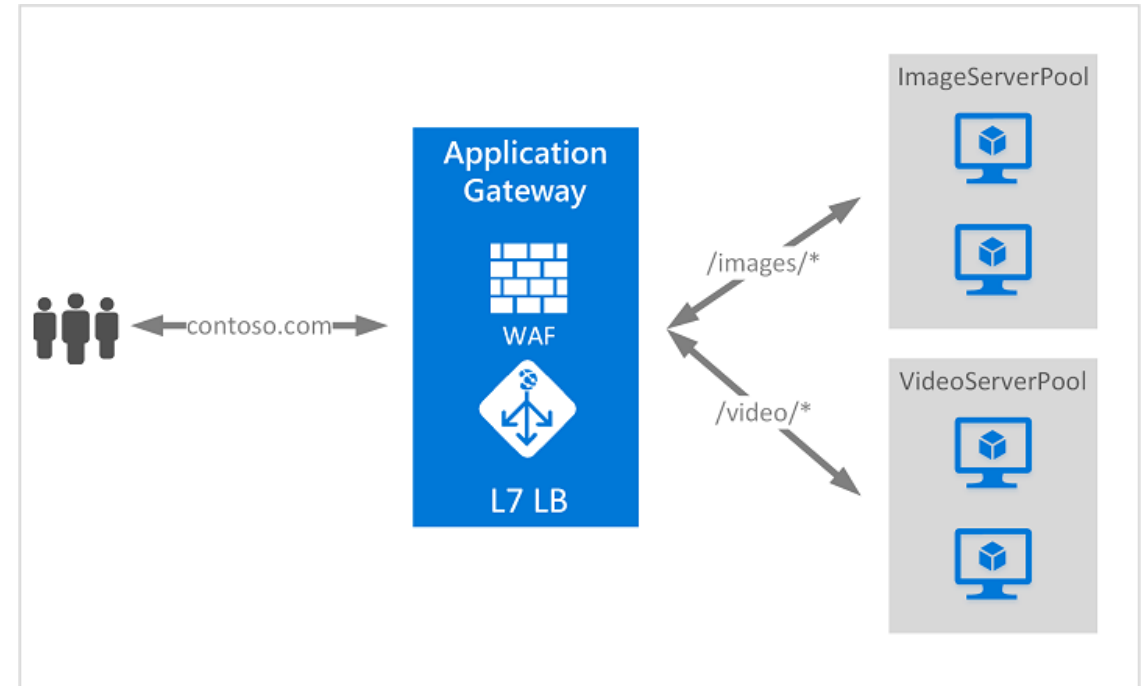
Load balancer

- Used to distribute evenly the incoming network traffic across a group of backend resources (VMs).
- Operate at OSI layer 4 – Transport (TCP/UDP).



Application gateway

- A web traffic load balancer.
- Can make routing decisions based on attributes of an HTTP request (OSI layer 7).



Demo

Using the Azure Portal, we will create the following resources:

1. VNet
2. App Service
3. Container Instance
4. Logic App

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Azure database services



- **Azure SQL Database:** relational database on the latest version of the Microsoft SQL Server database engine.



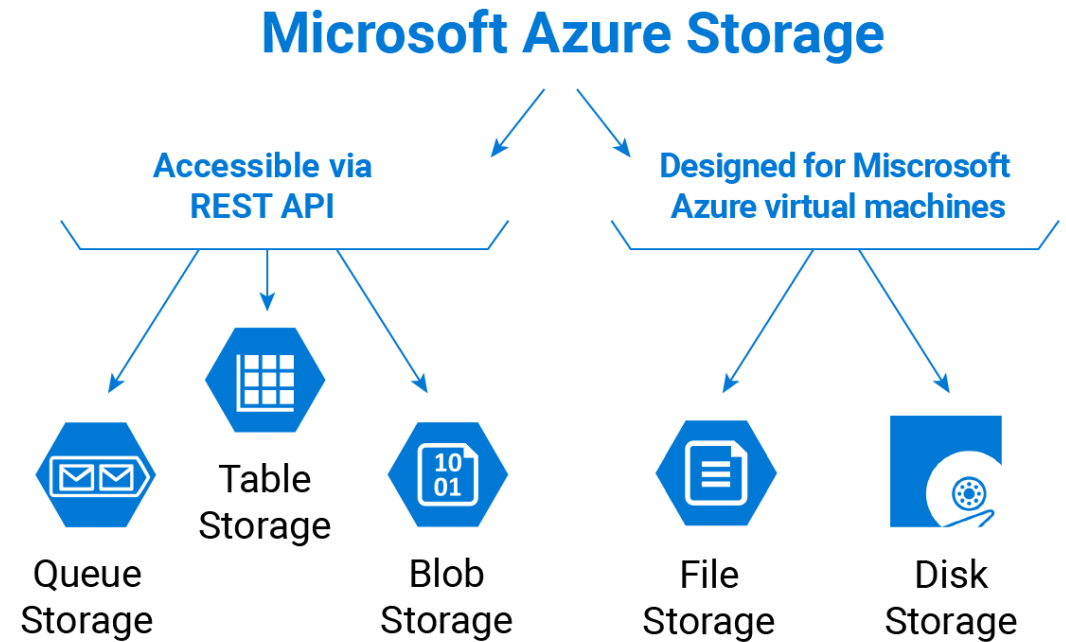
- **Azure CosmosDB:** a globally-distributed database service that enables you to elastically and independently scale throughput and storage.



- **Azure Database Migration:** a fully-managed service designed to enable seamless migrations from multiple database sources to Azure data platform with minimal downtime.

Azure storage account

- A central location for storing Objects, Files, NoSQL (also known as Tables and Queues) in a secure, scalable and redundant way.



Storage account types

Storage Account Type	Storage Service	Performance Tiers	Access Tiers	Replication Options	Encryption
General Purpose V2	Blob, File, Queue, Table, and Disk, and Data Lake Gen2	Standard, Premium	Hot, Cool, Archive	LRS, GRS, RA-GRS, ZRS, GZRS, RA-GZRS	Encrypted
General Purpose V1	Blob, File, Queue, Table, and Disk	Standard, Premium	N/A	LRS, GRS, RA-GRS	Encrypted
Block Blob Storage	Blob (block blobs and append blobs only)	Premium	N/A	LRS, ZRS	Encrypted
File Storage	Files only	Premium	N/A	LRS, ZRS	Encrypted
Blob Storage	Blob (block blobs and append blobs only)	Standard	Hot, Cool, Archive	LRS, GRS, RA-GRS	Encrypted

Access Tiers

Hot

- Intended for regularly accessed blobs.
- Low access rate charge.
- Most expensive storage charge per GB.

Cold

- Intended for rarely accessed blobs (under twice per month, maybe).
- Higher access rate charge.
- Lower storage charge per GB.

Archive

- Intended for data that probably will never be accessed anymore, but due to regulatory laws needs to be kept.
- For example, medical records.

Azure storage services

IaaS

Disks

- Persistent disks for Azure VMs.
- Disk options: HDD and SDD

Files

- For sharing files between users, replacing the File Server system
- Secure access from anywhere.

PaaS

Blobs

- Unstructured data such as images, videos.
- Ideal for serving content directly in a browser.

Tables

- It's a NoSQL storage service as a key attribute store for schemaless data.

Queues

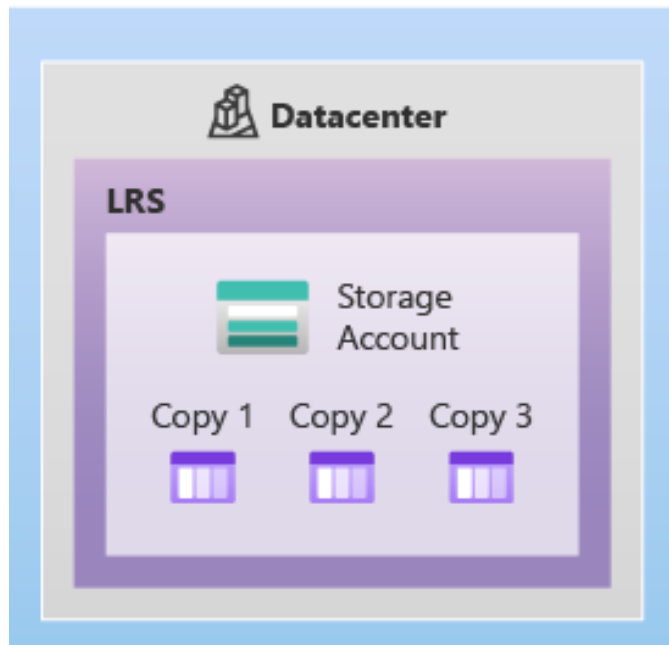
- Store and retrieve messages via HTTP/HTTPS calls to a Queue Name URL.

Replication in the primary region

LRS (Locally Redundant Storage)

- Data is replicated within 3 racks of storage nodes.

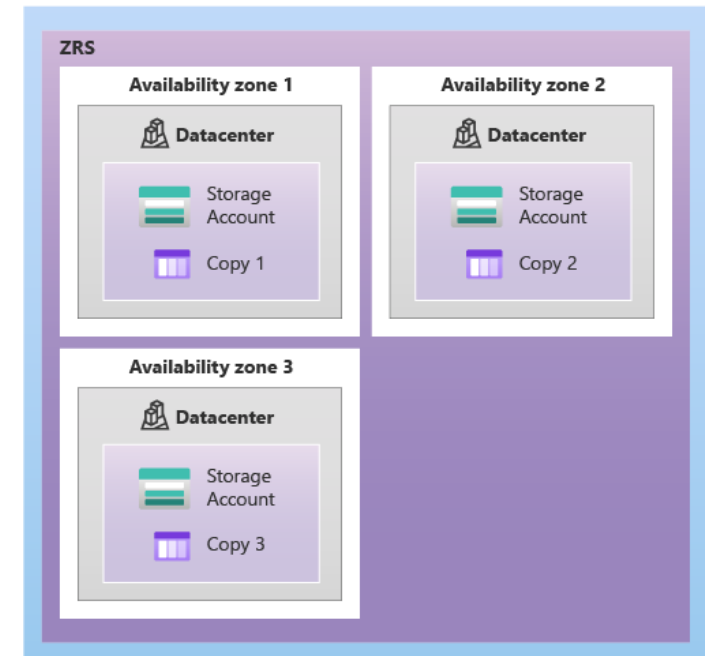
Primary region



ZRS (Zone Redundant Storage)

- Data is replicated over 3 data centers.
- If one data center is down, you can still access your data.

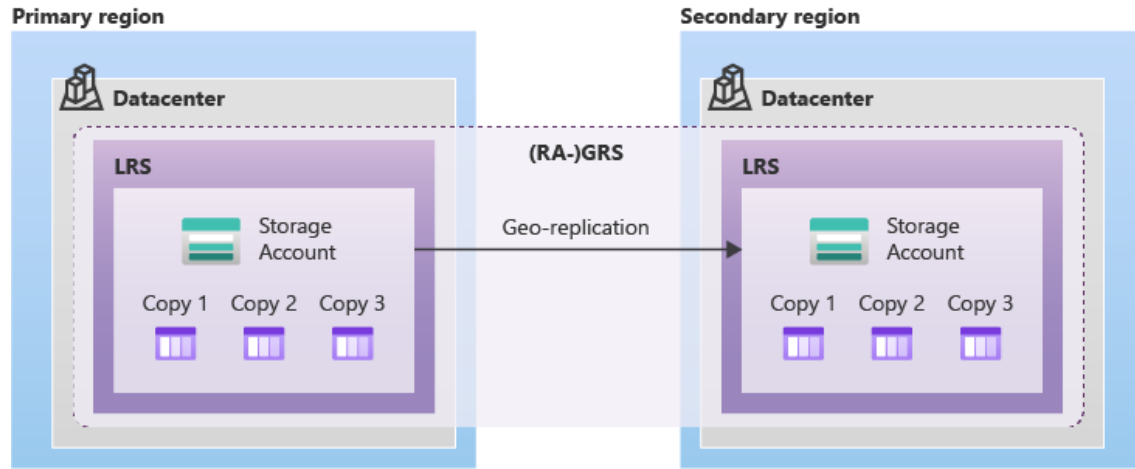
Primary region



Replication in a secondary region

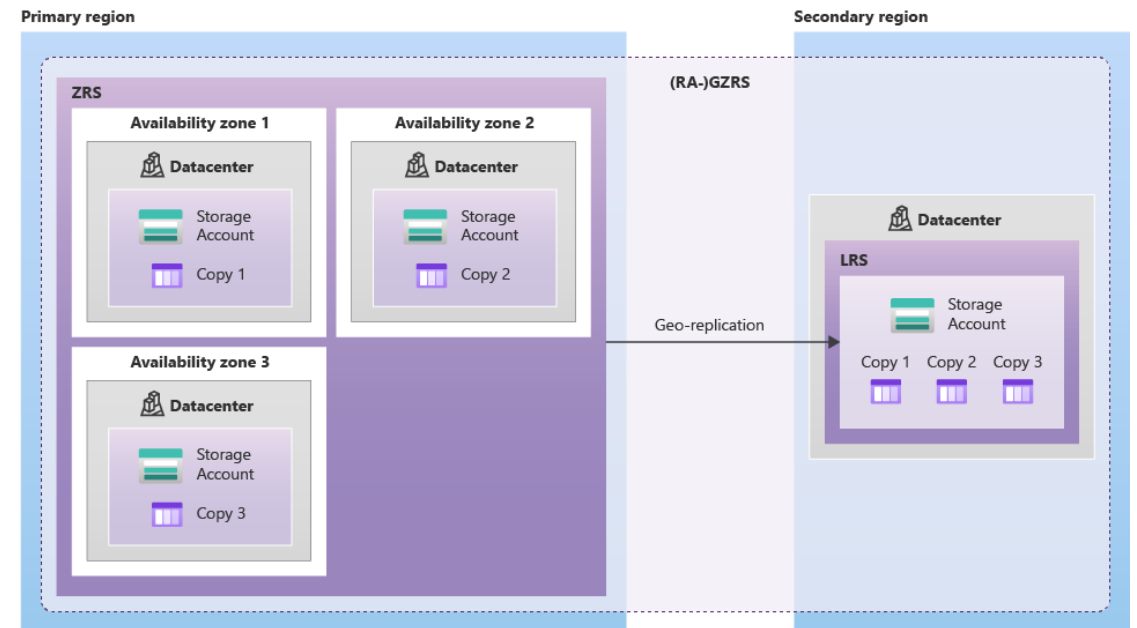
GRS (Geo-Redundant Storage)

- Data is replicated across another region.
- Replicated data is only available to read if only if Microsoft initiates a failover from the primary region.



GZRS (Geo-Zone Redundant Storage)

- Data is replicated over 3 availability zones in the primary region and copied asynchronously to a single physical location in the secondary region.



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Azure Monitor

- An all-in-one solution for your provisioned Azure Services or On-Premises.
- Gathers events, analyzes them and acts so you can understand how your applications are performing.



Metrics



Activity Logs



Diagnostic Logs

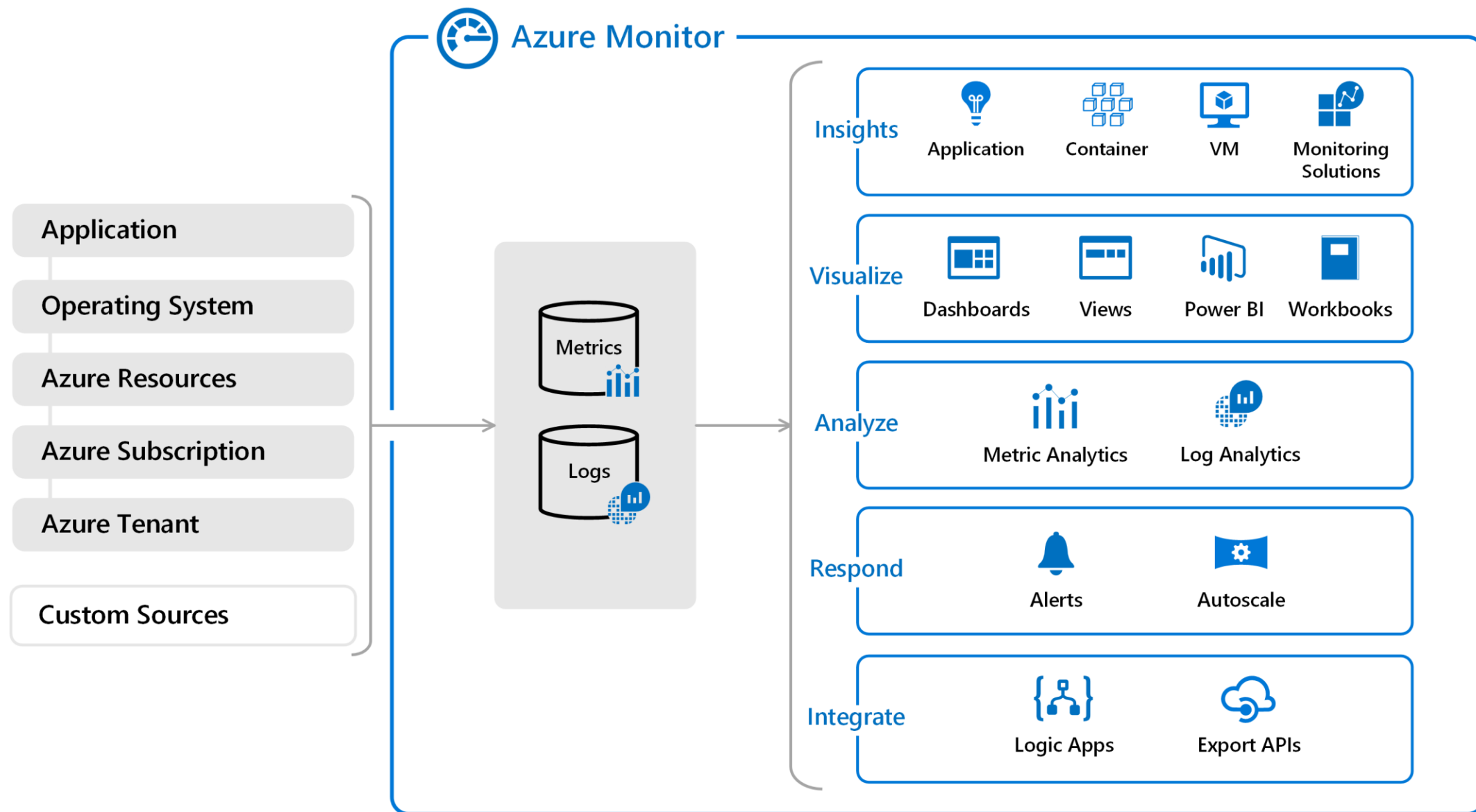


Alerts

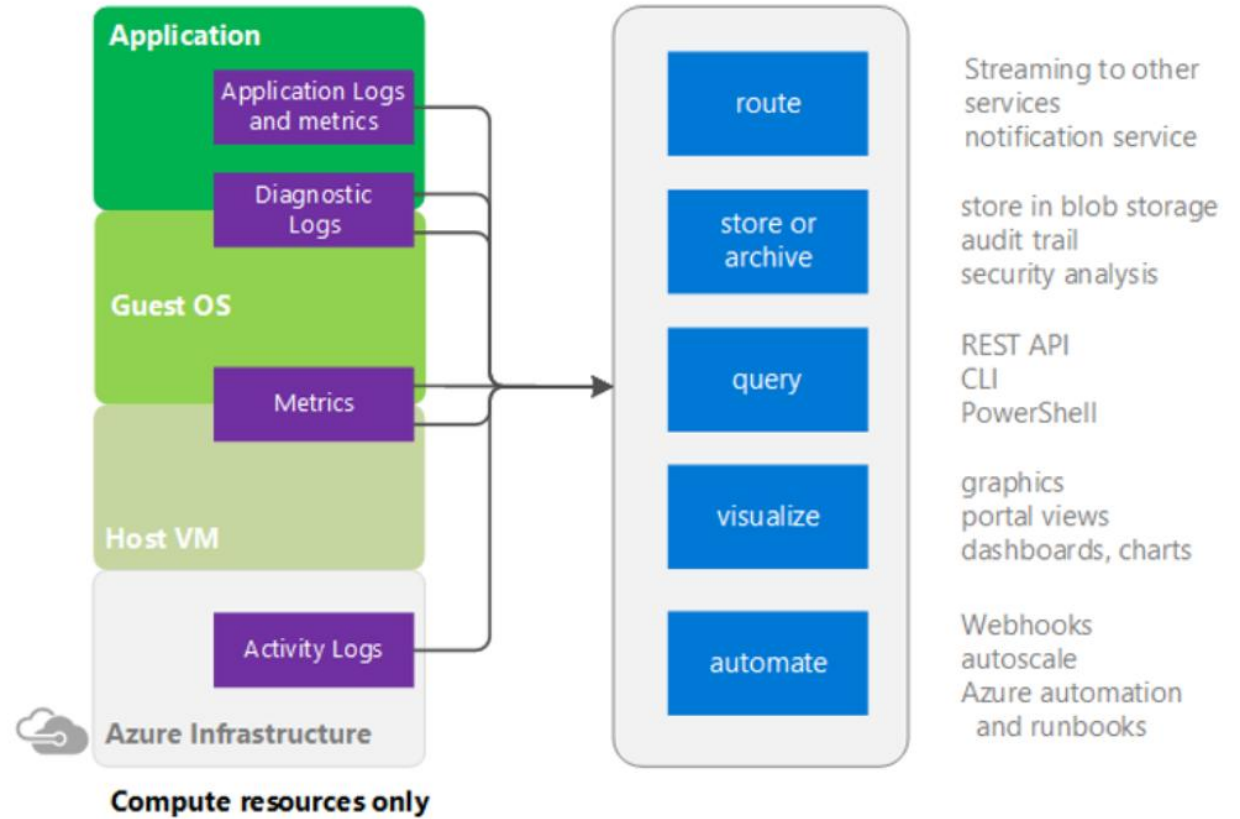
Health, metric & audit view
across all Azure resources



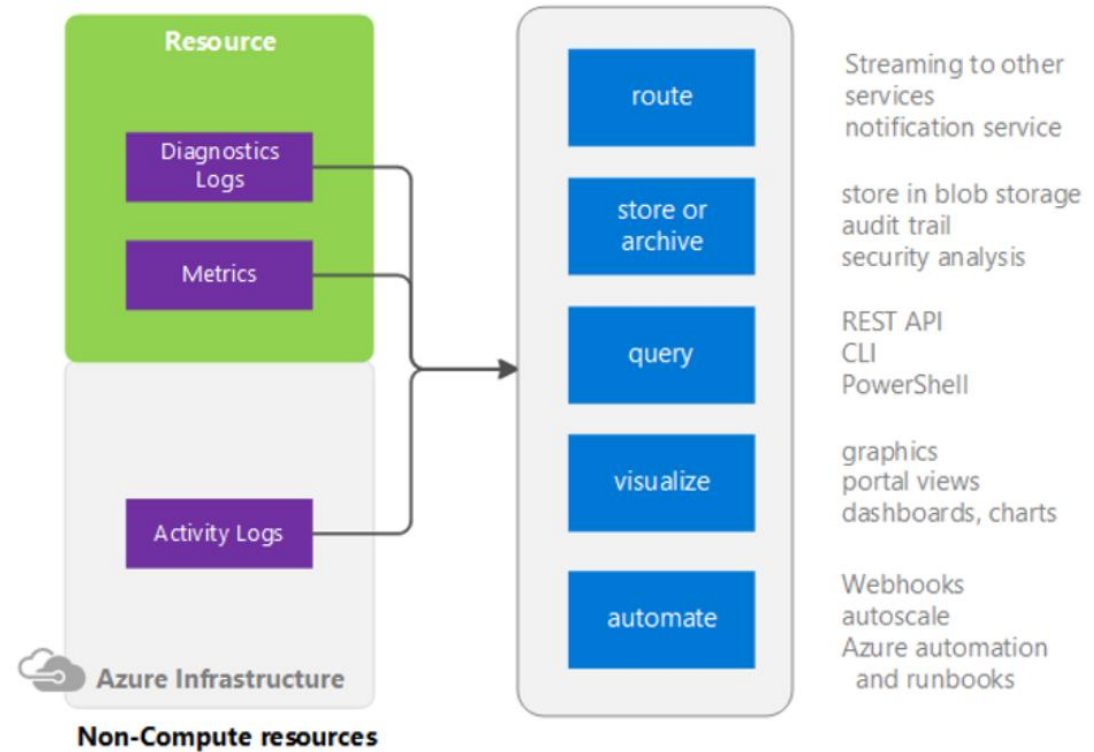
Azure Monitor



Azure monitor for compute resources

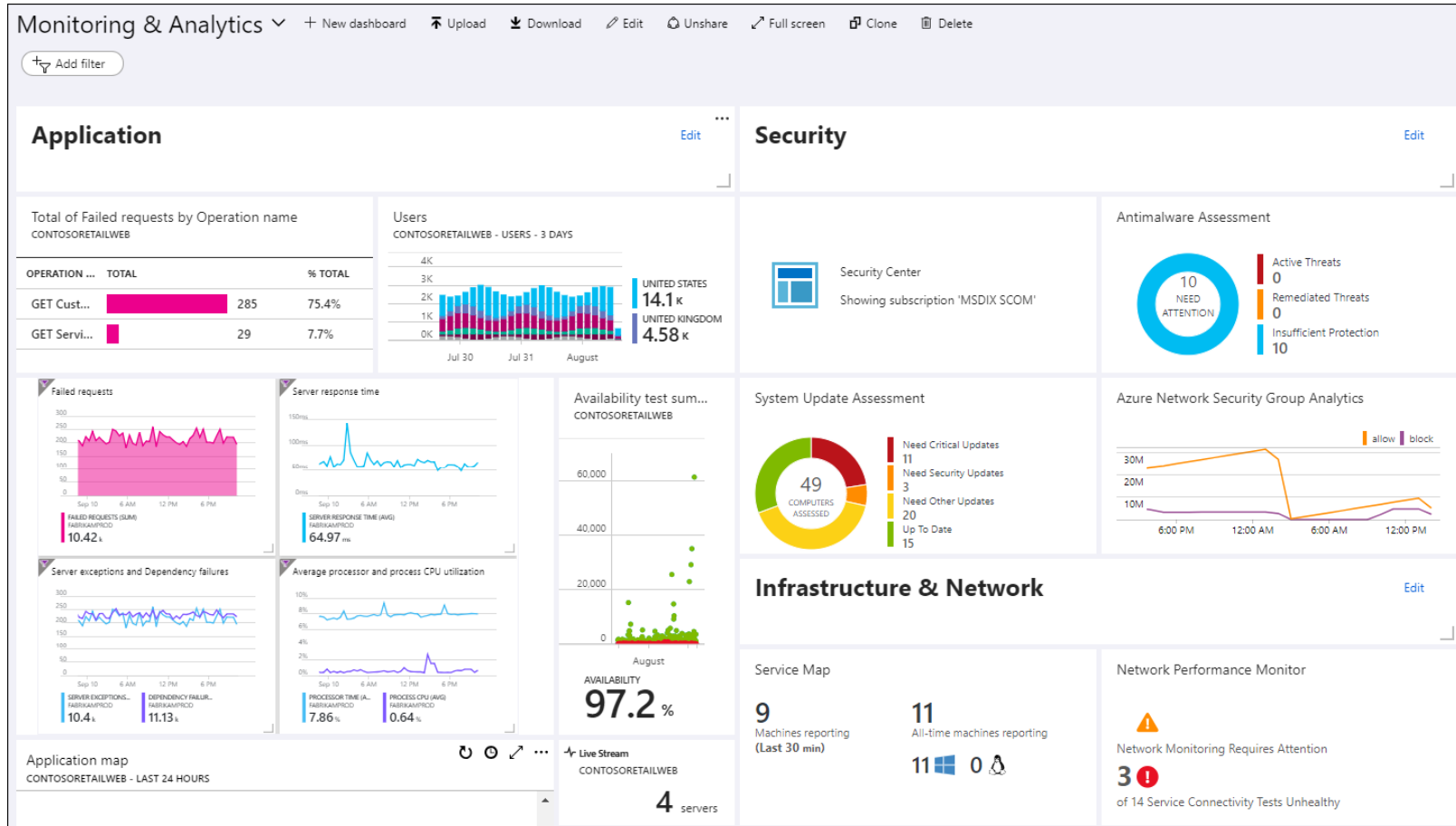


Azure monitor for non- compute resources





Azure Dashboards



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Shared security model

Responsibility	On-Premises	IaaS	PaaS	SaaS
Data governance and Rights Management	Customer	Customer	Customer	Customer
Client endpoints	Customer	Customer	Customer	Customer
Account and access management	Customer	Customer	Customer	Customer
Identity and directory infrastructure	Customer	Customer	Microsoft/ Customer	Microsoft/ Customer
Application	Customer	Customer	Microsoft/ Customer	Microsoft
Network controls	Customer	Customer	Microsoft/ Customer	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Azure security services



Security Center



Azure Advisor



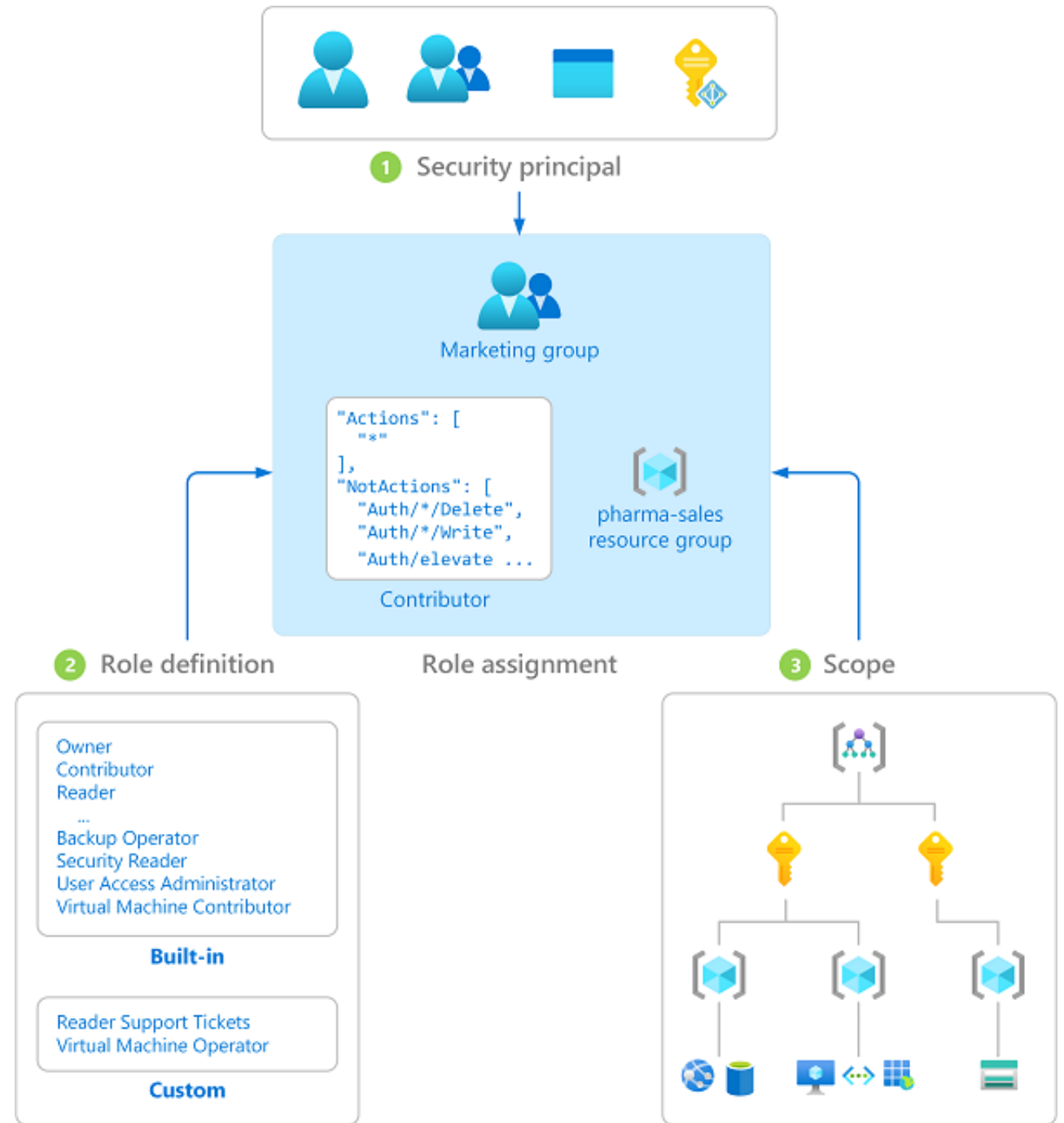
Network Security Group



Key Vault

RBAC

- Role-Based Access Control
- Access management for Azure resources.
- We can decide who has access to certain resources and what they can do with them.

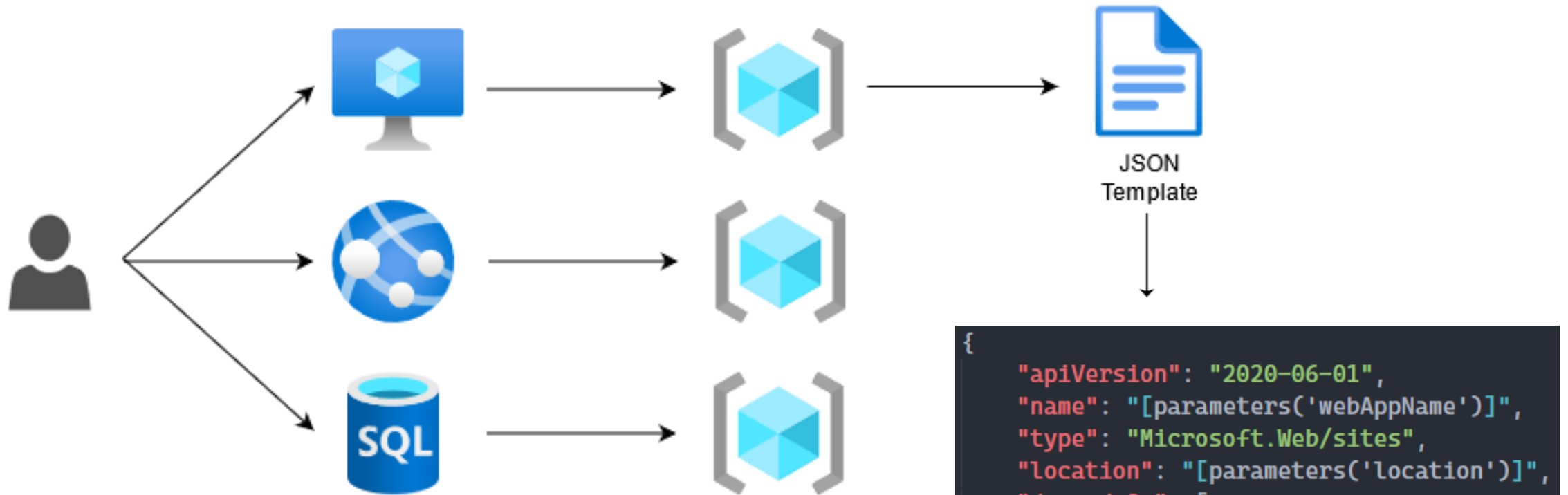


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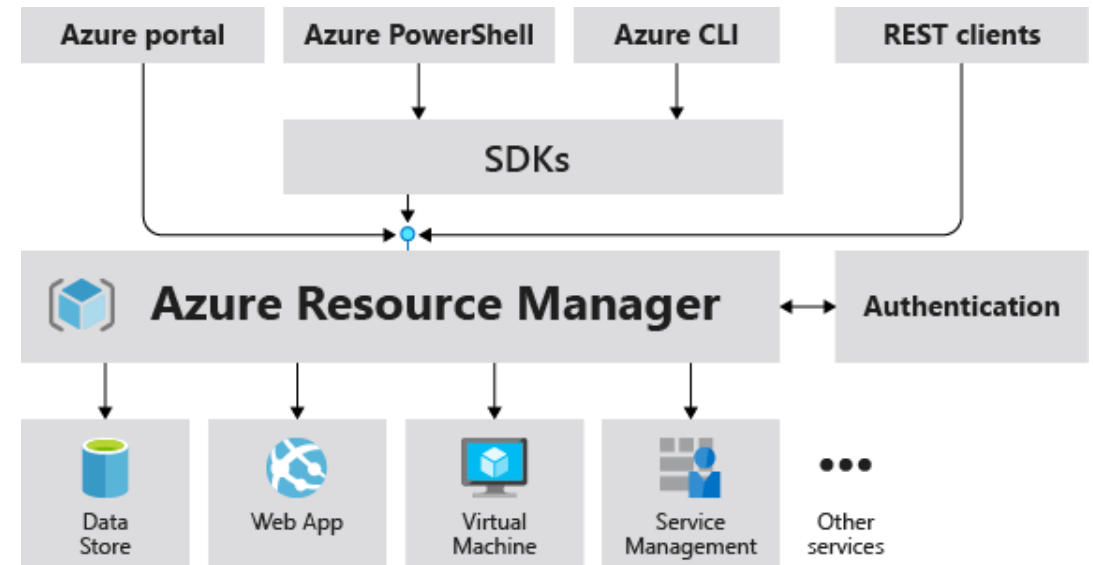


What are resources?



Introduction to ARM

- The deployment and management service for Azure.



What can you do with ARM?



Deploy



Organize



Control

Resource providers

- When you create a resource and a resource group in Azure you work with a set of Service Providers.
 - Each Service Providers expose a set of Resource Types which you create and manage.
- {resource-provider}/{resource-type}
 - Microsoft.KeyVault/vaults

Get-AzResourceProvider



Microsoft.Compute/
virtualMachines



Microsoft.Network/
publicIPAddresses



Microsoft.Storage/
storageAccounts



Microsoft.Network/
virtualNetworks



Microsoft.Network/
networkInterfaces



Microsoft.AppService/
apiapps

Imperative

Imperatively with
PowerShell or CLI

```
New-AzResourceGroup -Name $rgName `
                    -Location $locName
New-AzStorageAccount -Name $saName `
                    -ResourceGroupName $rgName `
                    -Type $saType `
                    -Location $locName
```

Declarative

```
"resources": [  
  {  
    "type": "Microsoft.Storage/storageAccounts",  
    "name": "[parameters('StorageAccountUniqueName')]",  
    "apiVersion": "2015-05-01-preview",  
    "location": "[resourceGroup().location]",  
    "properties": {  
      "accountType": "[variables('storageAccountType')]"  
    }  
  }  
]
```

Declaratively define
using a template

ARM Templates

"Create a new Azure resource with the following settings."



ARM Template

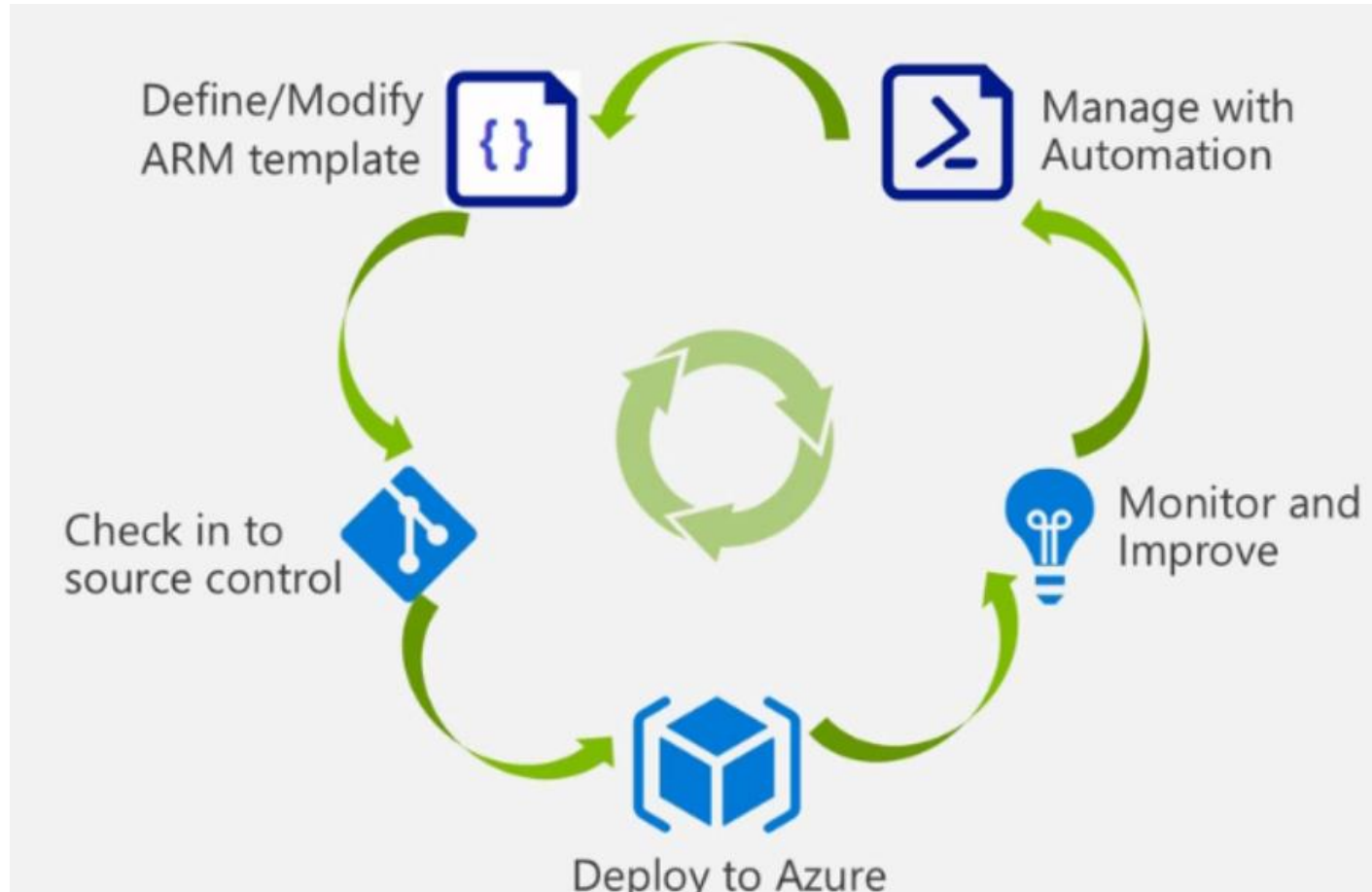


Azure



Resource Provisioning

App Lifecycle with Templates



Bicep Templates

- Domain Specific Language (DSL) for deploying Azure resources declaratively.
- All resource types, apiVersions, and properties that are valid in an ARM template are equally valid in Bicep on day one.




```
≡ vm_template.json
```

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#
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```

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301     privateLinkServiceNetworkPolicies: 'Enabled'
302   }
303 }
304
305 resource networkInterfaces_resource 'Microsoft.Network/networkInterfaces@2020-08-01' = {
306   location: 'westeurope'
307   name: networkInterfaces
308   properties: {
309     dnsSettings: {
310       dnsServers: []
311     }
312     enableAcceleratedNetworking: false
313     enableIPForwarding: false
314     ipConfigurations: [
315       {
316         name: 'ipconfig1'
317         properties: {
318           primary: true
319           privateIPAddress: '${VMIP}'
320           privateIPAddressVersion: 'IPv4'
321           privateIPAllocationMethod: 'Dynamic'
322           publicIPAddress: {
323             id: publicIPAddresses_resource.id
324           }
325           subnet: {
326             id: virtualNetworks_default.id
327           }
328         }
329       }
330     ]
331     networkSecurityGroup: {
332       id: networkSecurityGroups_resource.id
333     }
334   }
335   tags: {
336     env: tags_Env
337     service: tags_Service
338   }
339 }
340
```

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360     [resourceId('Microsoft.Network/virtualNetworks', parameters('virtualNetworks'))]
361   ]
362 },
363 {
364   "type": "Microsoft.Network/networkInterfaces",
365   "apiVersion": "2020-08-01",
366   "name": "[parameters('networkInterfaces')]",
367   "location": "westeurope",
368   "properties": {
369     "dnsSettings": {
370       "dnsServers": []
371     },
372     "enableAcceleratedNetworking": false,
373     "enableIPForwarding": false,
374     "ipConfigurations": [
375       {
376         "name": "ipconfig1",
377         "properties": {
378           "primary": true,
379           "privateIPAddress": "172.16.6.4",
380           "privateIPAddressVersion": "IPv4",
381           "privateIPAllocationMethod": "Dynamic",
382           "publicIPAddress": {
383             "id": "[resourceId('Microsoft.Network/publicIPAddresses', parameters('publicIPAdd
384           )]",
385           "subnet": {
386             "id": "[resourceId('Microsoft.Network/virtualNetworks/subnets', split(format('{0}',
387           )]"
388         }
389       ]
390     },
391     "networkSecurityGroup": {
392       "id": "[resourceId('Microsoft.Network/networkSecurityGroups', parameters('networkSecuri
393     }
394   },
395   "tags": {
396     "env": "[parameters('tags_Env')]",
397     "service": "[parameters('tags_Service')]"
398   },
399   "dependsOn": [
400     "[resourceId('Microsoft.Network/networkSecurityGroups', parameters('networkSecurityGroups
401     "[resourceId('Microsoft.Network/publicIPAddresses', parameters('publicIPAddresses'))]",
402     "[resourceId('Microsoft.Network/virtualNetworks/subnets', split(format('{0}/default', par
403   ]
404 }
405
```

Demo – Create an Azure Storage Account

- Using the Azure Portal create a Storage Account with a general purpose v2 container.
 1. Create a Storage Account
 2. Create a general purpose v2 container
 3. Using Azure Storage explorer, upload a file from your PC and verify that it was correctly uploaded.

Note: Storage Account name must be unique across all of Azure!

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